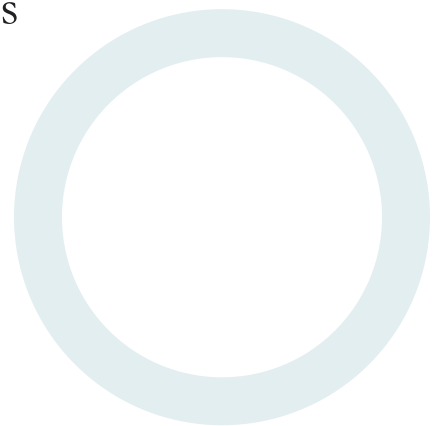
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Marketing Management

Week 4 · Managing Customer Dynamics

Marketing Principle #2: All Customers Change



Yixing Chen

Agenda

01 **Introduction**
Why customer needs and behavior change over time

02 **Lifecycle and dynamic segmentation**
Lifecycle logic, AER, hidden states, and migration

03 **Customer value and choice**
CLV, RFM, lost customers, and choice models

All Customers Change

- Another underlying issue facing managers that make marketing decisions difficult is that all customers change
- Customer's desires/needs for most products and services change over time or due to specific events
  - → Individual consumer needs change (age, experience, and due to trigger events)
  - → Customers are embedded in industries/markets, which change overtime (PCs 20 years ago and now)
- Customer's needs vary not only due to inherent differences in people (heterogeneity) but also as people and markets change (dynamics)
- Thus, we need to adapt our "static" segmentation of all customers based on "generic" needs (MP#1) by focusing on our existing customers and accounting for their time dependent needs (MP#2)
- **Customer dynamics** are changes in customer preferences that occur over time

Exercise on Customer Dynamics

- Everyone take a few minutes to describe your or your parents past car purchases
- Car at 20 _____
  - → Top two reasons bought _____ & _____
- Car at 30 _____
  - → Top two reasons bought _____ & _____
  - → Trigger for change _____
- Car now _____
  - → Top two reasons bought _____ & _____
  - → Trigger for change _____

Five Sources of Customer Dynamics: Individual Level

Dimension	Rate of change	Example
Discrete life events	**Immediate**	Becoming a parent may change preferences for cars, travel, housing, and restaurants.
Typical lifecycle or maturation	**Slow**	As people age, they may seek more comfort and health while becoming less willing to switch.
Product learning effects	**Medium**	Experience can increase demand for specialized features or make customers more price sensitive.

Five Sources of Customer Dynamics: Market and Environment

Dimension	Rate of change	Example
Product market level Product lifecycle	**Medium**	Early buyers may value novelty and performance. Later buyers may focus more on price, reliability, and compatibility.
Environmental level Economy, government, industry, and culture	**Slow to immediate**	Regulation, inflation, technology, or a cultural shift can rapidly change what customers value and how they buy.

Customer Dynamics: A Fundamental Assumption of Marketing Strategy

- Thus, customer dynamics is a fundamental "problem" that all firms must address when developing an effective marketing strategy
- Customers change; failure to understand and address these dynamics will lead to poor business performance

Marketing principle #2: all customers change and an effective marketing strategy must manage customer dynamics

Example: Nike vs. HOKA



NIKE, Inc.

FY2025 revenue

−10%



HOKA

FY2025 brand net sales

+23.6%

- Some runners increasingly seek cushioning, specialized performance, comfort, and distinctive design.
- Nike faced product and channel execution challenges as the running market became more specialized.
- HOKA built a clear position around enhanced cushioning, stability, and specialty running.

How should an incumbent respond when customer preferences fragment faster than its product portfolio and channels?

Sources: NIKE FY2025 full-year results; Deckers FY2025 results and Form 10-K. The revenue comparison does not establish that preference changes caused Nike's decline.

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Evolution of Approaches for Managing Customer Dynamics

- **Lifecycle Approach** uses generic stages of growth and their position in the lifecycle to determine customer preferences and associated strategies
- **Dynamic Customer Segmentation** segments a firm's existing customers on the basis of their similar, expected migration patterns
- **Customer Lifetime Value** captures the contribution of each customer according to his or her expected migration path over the entire lifetime with the firm

Evolution of Approaches for Managing Customer Dynamics

Slow — **Speed of Response** — Fast

All Customers — **Size of Segment Managed** — Niche Segment

Lifecycle Approach

Uses generic customer stages of growth and their position in the lifecycle to determine customer preferences and associated strategies.

- Customer lifecycle
- Product lifecycle
- Industry lifecycle

Pros	Cons
Simplicity	Assumes all customers follow one curve
Ease of use	Averages all customers
	Ignores causes of customer dynamics

Dynamic Customer Segmentation

Segments a firm's existing customers on the basis of their similar, expected migration patterns.

- May use hidden Markov models
- May use lost customer analysis

Pros	Cons
Combines lifecycle and segmentation methods	Segments are not perfectly homogeneous
Matches strategic marketing thinking	Puts continuous change into discrete stages
Identifies temporally homogeneous groups	

Customer Lifetime Value

Captures the contribution of each customer according to their expected migration path over the entire lifetime with the firm.

- Uses discounted cash flows
- Accounts for expansion and retention expectations

Pros	Cons
Provides insights for AER decisions	Requires insight into future migration
Supports a customer-centric culture	Requires detailed financial data
Captures dynamics and heterogeneity	

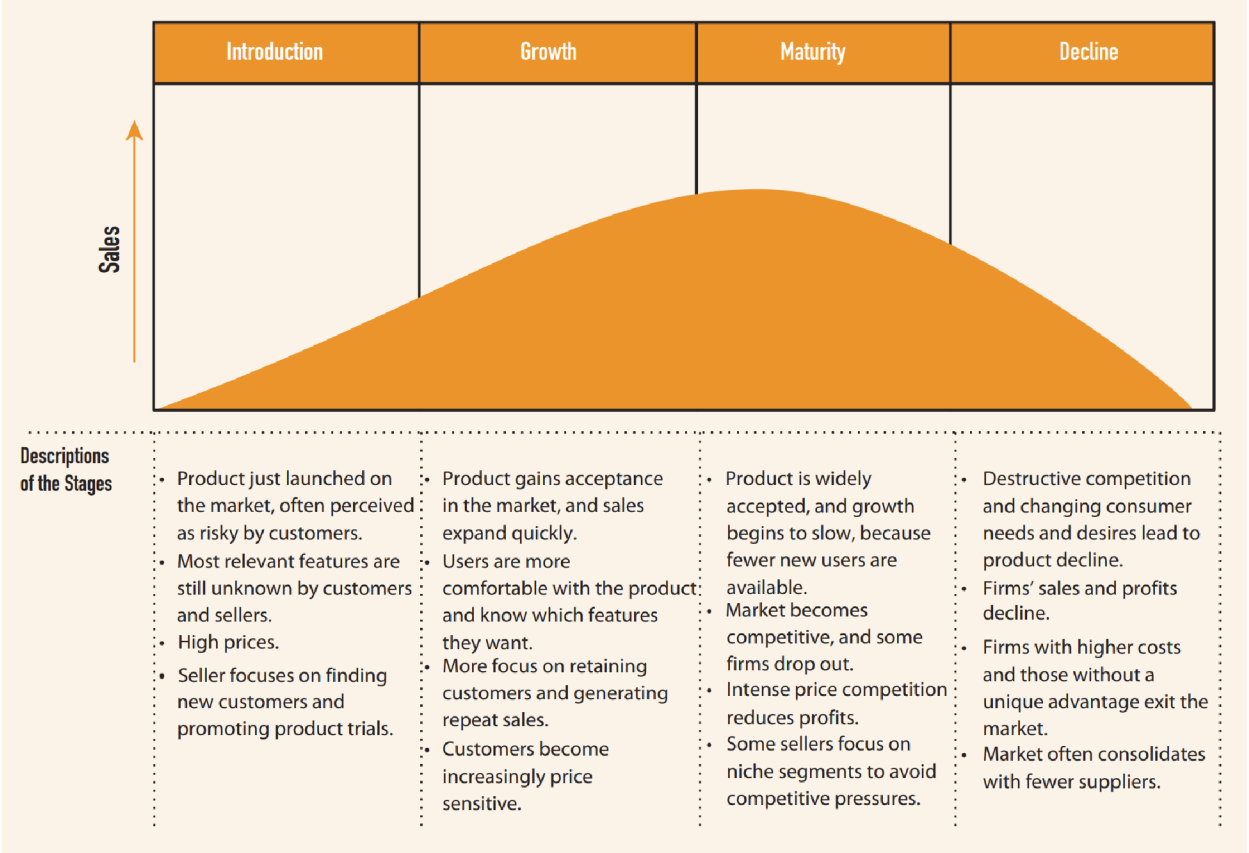
Lifecycle Perspective: "Ok" First Approximation

- Assumes "all" customers, markets, products, industries move thru a sequential set of "stages" as they mature
- Applies a standard set of marketing tactics at each stage
- Pros:
  - → Easy to apply
  - → Makes common sense
- Cons:
  - → Misses many unique trigger points and migration paths (ignores temporal heterogeneity)
  - → Inaccurate and/or ineffective for many firms

Different Lifecycle Approaches

- **Customer lifecycle** attempts to capture how individuals typically change as they age and reach common age-related milestones
- **Product lifecycle** proposes that various products go through four typical stages in relation to their acceptance by society: introduction, growth, maturity, and decline
- **Industry lifecycle** comprises of five stages:
  - → Early establishment of its range and boundaries
  - → An innovation stage to set a "dominant design"
  - → The shakeout stage, marked by economies of scale, such as that smaller players get forced out
  - → Maturity, when firms focus on market share and cash flows
  - → The decline stage, when sales decay for the industry as a whole

Typical Customer Product Lifecycle



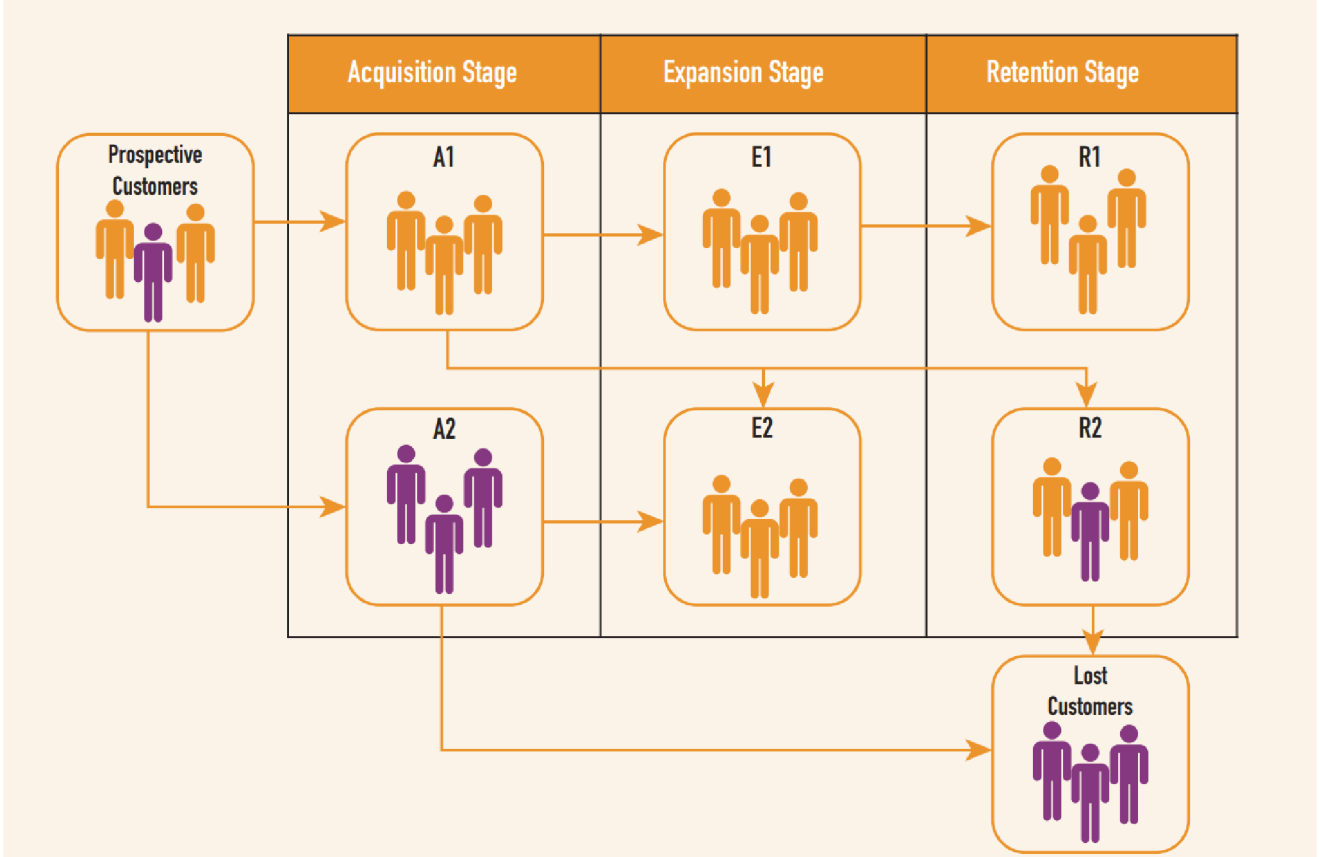
Example: Funai (Japan)

- The VCR was developed in the 1950s, and the United States alone housed over 10 million VCR units during the product's peak in the 1980s.
- Decades after the VCR's introduction, households switched to DVDs and then streaming services.
- Funai Corporation struggled to maintain business during the decline stage of the VCR's product lifecycle.
- In 2016, Funai, the last known manufacturer of VCR devices, ceased production, citing the difficulty of acquiring parts and the lack of sales.
- Customers' changing preferences and new product introductions sounded the death knell of the VCR and Funai's production of the device.

Customer Dynamic Segmentation Approach

- Evaluates **existing customer's** behaviors/needs in each AER stage to understand temporal differences
  - → Customers are "temporally" similar in each stage (assumption)
  - → Matches marketing action domains (i.e., acquisition is often a self-contained marketing domain)
- Dynamic-based segmentation is sometimes called the ***Acquisition–Expansion–Retention (AER) model***, because it captures customers entering the firm's portfolio and expanding over time, even as other customers slowly leave
  - → **Acquisition stage** begins with first contact, typically before the first purchase occurs, when prospects and early customers have similar needs
  - → **Expansion stage** has firms that are trying to upsell or cross-sell to expand their sales and engagement with existing customers
  - → **Retention stage** deals with customers who migrate not because of a mismatch in the core offering or a life event but because they have a basic propensity to switch, in pursuit of "greener pastures"

Customer Dynamic Segmentation Approach (AER Model)



Hidden Markov Models and Dynamic Segmentation

A customer's relationship with a firm can occupy an unobserved state that changes over time. Purchases, engagement, and marketing responses provide signals about that state.

- Customers in the same state display similar behavior.
- Each customer has a probability of moving to another state.
- Marketing actions may influence those transition probabilities.



B2B Relationship Case

Business problem

A supplier wants to understand why apparently stable customers strengthen, weaken, or damage their relationships over time.

Longitudinal data

346 B2B customers observed over six years, including trust, commitment, dependence, relational norms, cooperation, and profit.

Analytical questions

- Which hidden relationship states exist?
- How do customers migrate among them?
- Which actions encourage favorable migration?

Hidden Markov Model Workflow

1
Organize panel data

Build a customer-by-period record of behavior, outcomes, and marketing actions.

2
Estimate hidden states

Compare solutions with different numbers of states and interpret the behavioral profile of each state.

3
Estimate migration

Calculate initial and transition probabilities for moving from one state to another.

4
Connect actions to change

Test whether marketing activity changes the probability of favorable or unfavorable transitions.

What the Hidden Markov Model Estimates

1. **Number of states**How many distinct relationship patterns the data support
2. **Initial state probabilities**Where a customer is likely to begin
3. **Transition probabilities**How likely customers are to move between states
4. **Behavior within each state**The expected purchases, engagement, or outcomes conditional on state
5. **Marketing effects**How firm actions change the likelihood of migration

Results: Four Relationship States

Transactional

Low trust, commitment, dependence, and relational norms.

Transitional

A short-lived bridge as customers explore a stronger relationship.

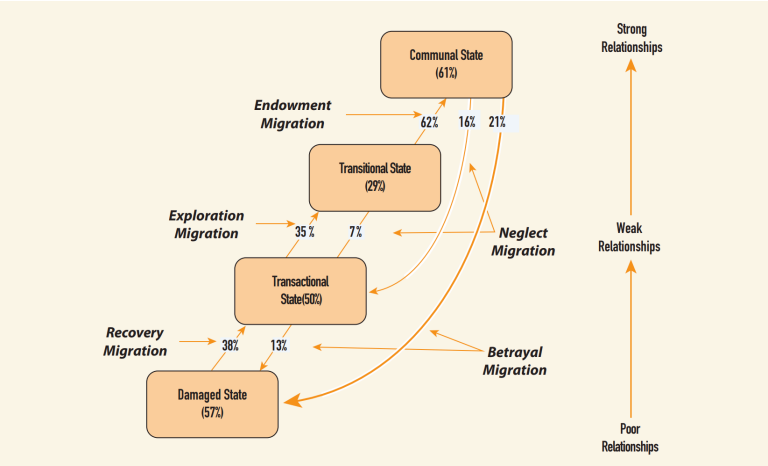
Communal

High trust and commitment, strong cooperation, and the highest profit.

Damaged

Low trust and cooperation, but customer dependence can remain high.

Results: Relationship States and Migration Paths



What the paths reveal

- Customers can move in either direction between strong and weak relationships.
- Positive migration often passes briefly through the transitional state.
- Dependence alone does not prevent a relationship from becoming damaged.

Insights from the B2B Hidden Markov Model

- A state describes behavior at one point in a relationship. It does not permanently label the customer.
- Migration probabilities make customer dynamics measurable rather than anecdotal.
- Brief transitional states can provide an early window for relationship building.
- High dependence may coexist with low trust, so retained revenue can hide relationship risk.

Which customer signals would tell a manager that a relationship is moving toward a communal or damaged state?

Customer Lifetime Value (CLV) is a Key Analysis Tool for Making AER Decisions

- How do you know best customers to acquire/expand/retain?
  - → In many banking initiatives only 1 in 3 "customers" remain after incentive ends
  - → Are all customers worth acquiring or retaining?
- **CLV approach:** evaluates a firm's profit as the sum of each customer's lifetime discounted cash flows
- Approach captures "true" contribution of each customer at any stage by accounting for:
  - → **Customer heterogeneity** and **dynamic effects** (individual level, uses transition expectations, and discounts future profits)
  - → **Tradeoffs among AER strategies** (e.g., how acquisition may affect retention)

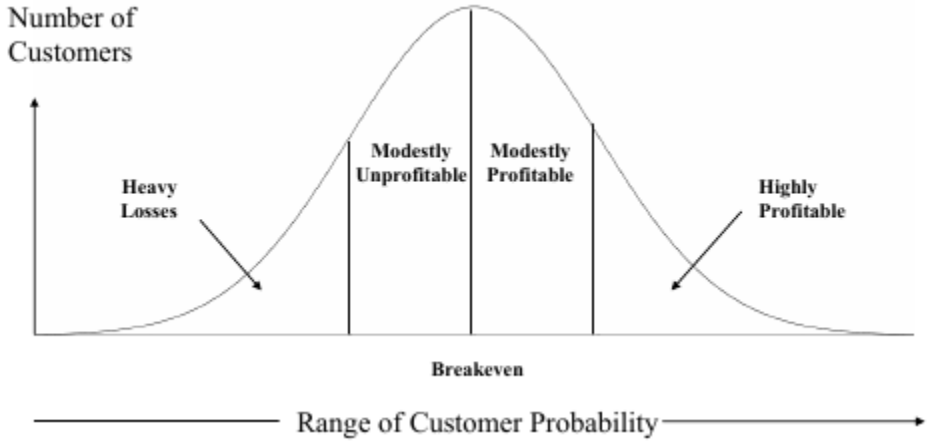
CLV Approach (Australia and New Zealand CMO Survey)

- Survey conducted on 255 CMOs and marketing directors in Australia and New Zealand
- Those who "always measure the lifetime value of each customer" achieved a 16% average increase in their annual marketing budget as compared to 0% for those who do not measure it.
- 75% of the marketers are engaged in some level of CLV effort within the organization



CLV Accounts for Varying Profits Across Customers

Beyond 80/20 rule: firms earn 150% of their profits from 30% of their customers
CLV captures these difference in your existing customers so you can acquire, expand, and retain the "best customers"



The Right Customers: Acquisition, Retention, and Development (HBR Press)

CLV Accounts for the Time Varying Profits of Your Customers

On average, annual earnings typically increase over a customer's life due to cross/up-selling
But, some customers are more costly to acquire or retain (lowering prices, high service levels)

	Profit per Customer (in dollars) by Year of Relationship				
Industry	1	2	3	4	5
Credit Card Issuance and Servicing	(21)*	42	44	49	55
Industrial Laundry	144	166	192	222	256
Industrial Distribution	45	99	121	144	168
Auto Servicing	25	35	70	88	88

* Figures in parentheses denote losses.

The Right Customers: Acquisition, Retention, and Development (HBR Press)

Customer Lifetime Value Analysis

Customer lifetime value measures the present value of expected customer cash flows over the relationship.

$$CLV_i = \Sigma_{t=0}^{T} CF_{i,t} / (1 + d)^t$$

- **$CF_{i,t}$** is the net cash flow generated by customer i in period t.
- **T** is the time horizon.
- **d** is the discount rate.
- Firm value can be viewed as the sum of customer lifetime values.

Simplified Customer Lifetime Value

$$CLV_i = (M_i - C_i) / (1 - r_i + d) - A_i$$

M_i Annual margin

C_i Annual marketing cost

r_i Retention rate

d Discount rate

A_i Acquisition cost

Assumes stable annual margin, marketing cost, and retention over a long horizon.

Customer Lifetime Value: Purpose and Use

Description

CLV converts expected future customer profit into a present value at the customer or cohort level.

When to use it

- Identify customers worth acquiring, expanding, or retaining.
- Compare marketing investments with expected customer value.
- Support customer-centered resource allocation.

How it works

Estimate the relationship horizon, forecast net cash flows, discount future values, and subtract acquisition cost.

Interpreting the Simplified CLV Formula

$$CLV_i = (M_i - C_i) / (1 - r_i + d) - A_i$$

Higher margin raises CLV

Higher retention raises the expected relationship value

Higher discount rate reduces the value of future cash flows

Higher acquisition cost reduces CLV directly

CLV Example: The Brett Family

Inputs

Annual margin	$100
Annual marketing cost	$10
Retention rate	65%
Discount rate	5%
Acquisition cost	$65

Calculation

($100 − $10) / (1 − 0.65 + 0.05) − $65

CLV = $160 The positive value supports acquisition, subject to uncertainty and comparison with alternative customers.

Blue Apron: Growth Versus Customer Economics

Blue Apron led the U.S. meal kit market while the category grew quickly. Customer metrics, however, pointed to a weaker economic model.

- The company kept adding customers as acquisition cost increased.
- Later customer cohorts generated less revenue on average than earlier cohorts.
- Sales growth could therefore coexist with declining value per acquired customer.

53%

U.S. meal kit market share reported for 2016

Rising CAC

Customer acquisition became more expensive

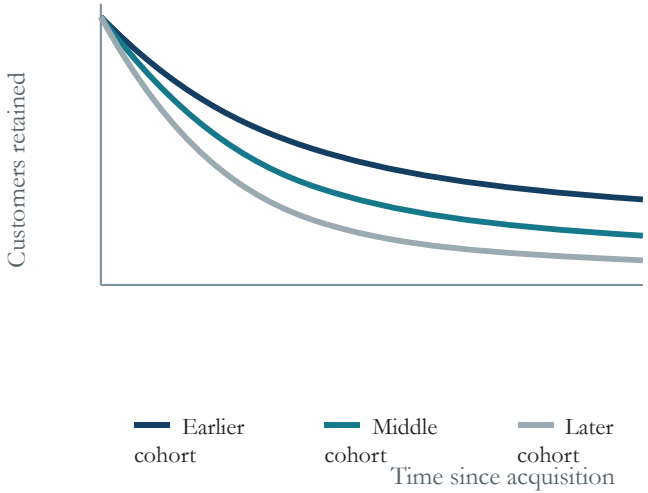
Lower cohort revenue

Newer customers produced less revenue on average

Should management keep pursuing customer and revenue growth, or improve retention, acquisition quality, and CLV?

Source: McCarthy and Schwartz, Blue Apron: Turning Around the Struggling Meal Kit Market Leader, WDI Publishing, 2018. Case product W73C09.

Blue Apron: Retention Curves and Cohort Quality



What the curve reveals

- The initial drop shows how many acquired customers fail to form a relationship.
- The long tail determines how many future orders and margins remain.
- Cohort comparisons separate growth in customer count from improvement in customer quality.

The curves are schematic. Use the case spreadsheet to reproduce the published estimates and compare segments.

Blue Apron: CLV, CAC, and Acquisition ROI

Retention curve

Expected survival by period

Future contribution

Orders × contribution margin

Customer value

Discounted future contribution

Acquisition ROI

Compare CLV with CAC

CLV comfortably exceeds CAC

Continue acquiring similar customers while monitoring cohort quality.

CLV is close to CAC

Improve targeting, offer design, and early retention before scaling.

CLV falls below CAC

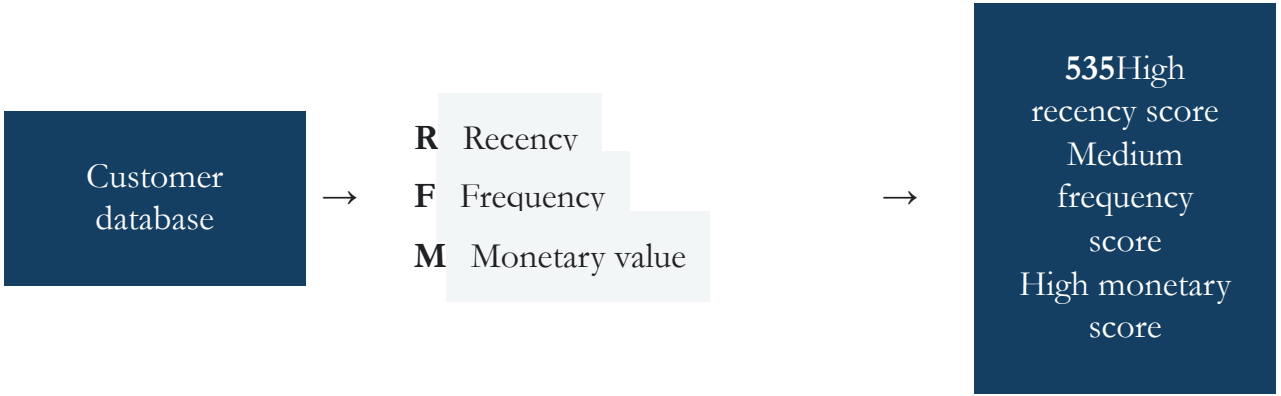
Reduce acquisition spending for that segment or redesign the customer experience.

Case, spreadsheet, and instructor-material access Access depends on the distributor and educator account.

RFM Analysis is "Poor Man's" CLV

- Direct marketers have been using a simplified version of the CLV for decades, targeting customers to receive expensive catalog mailings. They use three readily available customer behaviors:
  - → *Recency* or time elapsed since last purchase
  - → *Frequency* of purchases in last period
  - → *Monetary* purchases in last period
- These RFM (recency, frequency, and monetary) variables put customers in rank-ordered groups, based on their value in the past year (not by modeling but by rank-order sorting)
- Using the profits generated from a test mailing to a few customers from each group, direct marketers then mail the catalog only to the groups with an acceptable return on investment

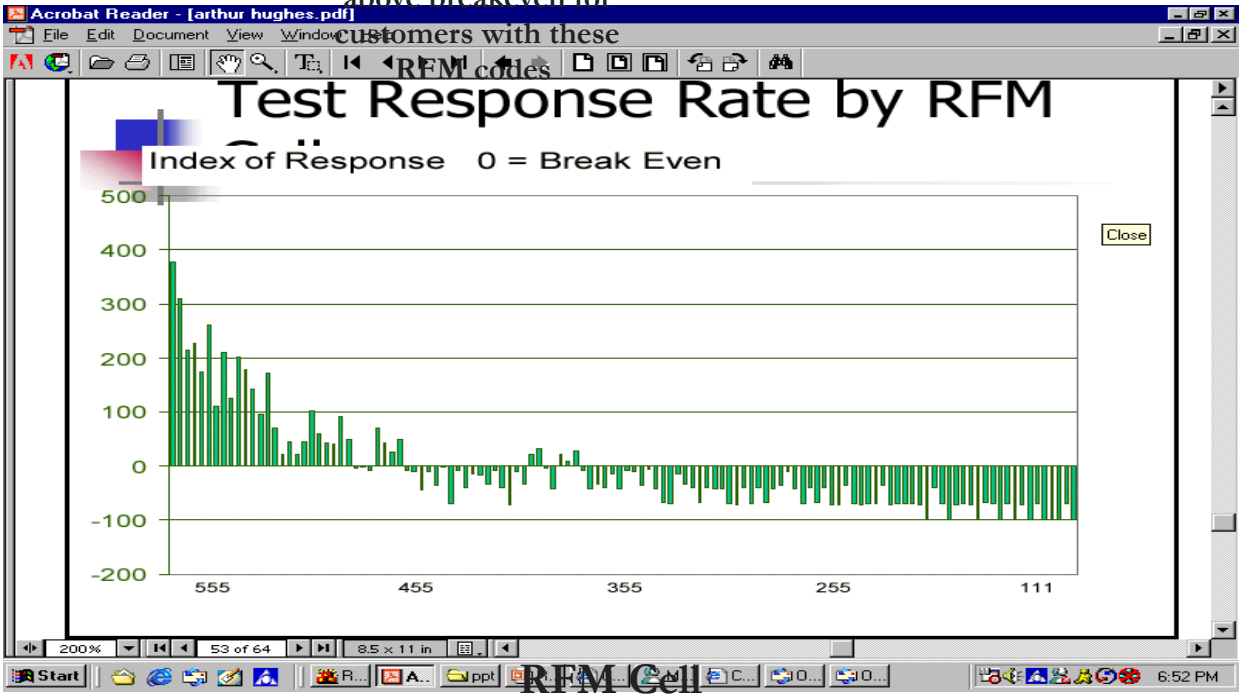
RFM Code Construction



> Higher scores indicate more recent, frequent, and valuable purchasing behavior.

Test of a Expansion Strategy Using RFM Analysis for Mailing

Catalog mailing is above breakeven for customers with these RFM codes



Breakeven Analysis

RFM Cell

Lost Customer Analysis Informs AER Strategies

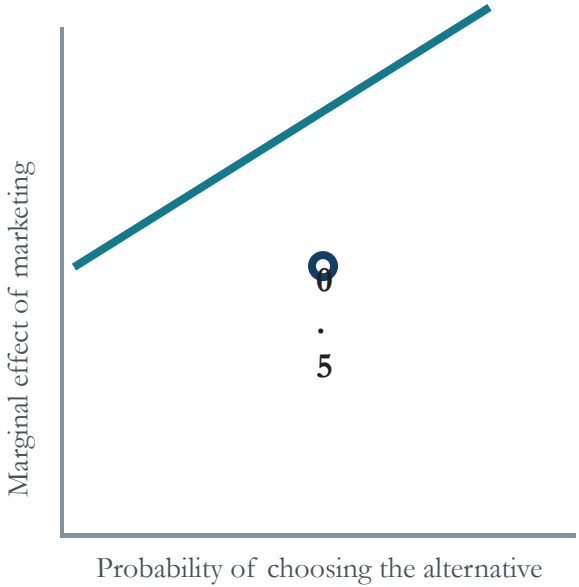
- A firm contacts customers that have migrated away, to identify the cause for this change, then works backward to fix the problem and ensure other customers don't leave for the same reason
- Takes a significant number of lost customers before a firm recognizes that it isn't just "normal" customer churn but rather an indication of an underlying problem
- **Three-step process**, which provides insights into both strengths and weaknesses:
  - → Firms set regular intervals for contacting lost customers to identify the cause of their transition, where they went, and potential recovery strategies
  - → If the lost customer is not in the firm's main target segment, firms could:
  - → Change their acquisition criteria
  - → Evaluate an expansion strategy to address a new subsegment of customers
  - → If the lost customer is in the firm's target market, firms should:
  - → Fix the problem
  - → Implement retention strategies to build brand and relational loyalty
- A **choice model** can benefit the lost customer analysis, and can inform analyses across all AER stages, because it predicts the likelihood of observed customer choices/responses, using data about that customer's characteristics and past behaviors, as well as the firm's marketing interventions
-

What are Choice Models?

- Analysis approach that attempts to determine the impact of different factors (price, promotion) on consumer's individual choices (joining, cross buying, leaving)
  - → Most popular individual-level response model
  - → Uses past behavioral data
  - → No need to survey or get customer input (infers weights from past customer's behaviors)
  - → Determines probability of choices and elasticities of marketing variables
- While regression analysis…
  - → Is not an individual model and uses continuous outcomes
  - → Gives weights of inputs, but not probabilities or elasticities
  - → Does not use "logit model," which is better predictor of how people act by capturing diminishing effects (inputs have less effect as you do it more)

(See Data Analytic Technique 3.3 for more details)

Choice Models and Diminishing Sensitivity



- Choice models assume that customers select the option with the highest individual utility.
- Marketing has the greatest marginal effect for customers who are close to indifferent.
- Additional effort has less effect when the choice probability is already near zero or one.

Choice Models are Excellent for Determining "Best" AER Strategies

- After determining AER positioning for each stage and ranking personas on CLV, then you need to develop effective AER marketing strategies
- Use database of past marketing actions and demographics linked to actual customer responses (choices) in a stage as input to choice model
- Choice model will provide:
  - → Elasticity for every input variable on outcome (e.g., how does age, kids, credit, and direct mail impact choice)
  - → **Elasticity** = % change in output / % change in input
  - → **Probabilities of customer's choice** (probability of upselling, retaining, and can run on lists for acquisition targeting)
- **Latent class choice models**: "clusters" and runs choice model at the same time (segments on elasticity to IVs, same things work for segment)

Choice Model Example: MBA Program Inputs

Observed choice

Whether an applicant enrolled in the program.

Predictors

- Family income
- Intensity of contact with the EMBA office
- Distance from home to campus
- Coworkers already in the program

Applicant	Enroll	Income	Contact	Distance
1	1	3	7	1
2	0	6	4	7
3	0	6	4	7
4	1	7	7	1

Choice Model Outputs

Coefficient estimates

Direction and strength of each predictor's association with enrollment.

Predicted probabilities

Enrollment likelihood for each applicant under observed or simulated conditions.

Elasticities

Expected percentage change in choice probability for a percentage change in an input.

Use holdout accuracy and managerial plausibility to judge whether the model is useful.

Choice Model Analysis: Purpose and Use

Description

A choice model estimates how customer attributes and marketing actions influence the probability of an observed choice.

When to use it

- Predict which option a customer will choose.
- Estimate the importance of price, promotion, and customer characteristics.
- Segment or target customers by response.

How Choice Models Work

1
Represent each option with a customer-specific utility.

2
Link the observed choice to option attributes and customer characteristics.

3
Estimate coefficients and convert utilities into choice probabilities.

4
Use probabilities to predict behavior and simulate market share under alternative actions.

Choice Model Example: Store-Brand Bleach

Problem

A retailer considers introducing Brand C into a category dominated by Brands A and B.

Data

Transactions record sales, prices, and promotional intensity for the two incumbent brands.

Model insights

- Lower price increases a brand's share but can shift demand across all alternatives.
- More frequent promotion raises sales, although the effect differs by brand.
- The retailer introduces Brand C as a lower-price option supported by frequent promotions.

Full data and model diagnostics are required before treating these elasticities as precise forecasts.